School of Electrical Engineering & Telecommunications

ELEC1111 Workshop 4

Q1 (2020 T3 Midterm Exam Q5)

Find the voltage across the capacitors in the circuit of Figure 1 under DC steady state conditions.

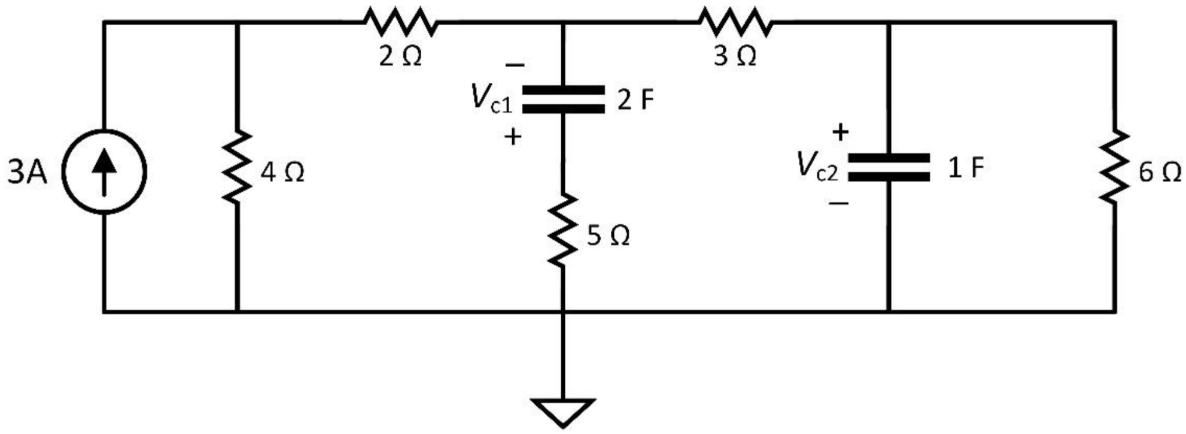


Figure 1

Answers: V_{c1} = -7.2 V, V_{c2} = 4.8 V

Q2 (Modified from 2022 T3 Midterm Exam Q5)

The switch in the circuit shown in Figure 2 was open for a long time. It closed at t = 0s and after remaining closed for 5 ms, it opened again.

i) Calculate $v_c(t)$ for 0 ≤ t < 5 ms.

ii) Calculate $v_c(t)$ for t ≥ 5 ms.

iii) Calculate the value of $i_c(\mathrm{t})$ at t = 10 ms.

iv) Determine the current flowing through the 20 Ω resistor for the time interval 0 ≤ t < 5 ms.

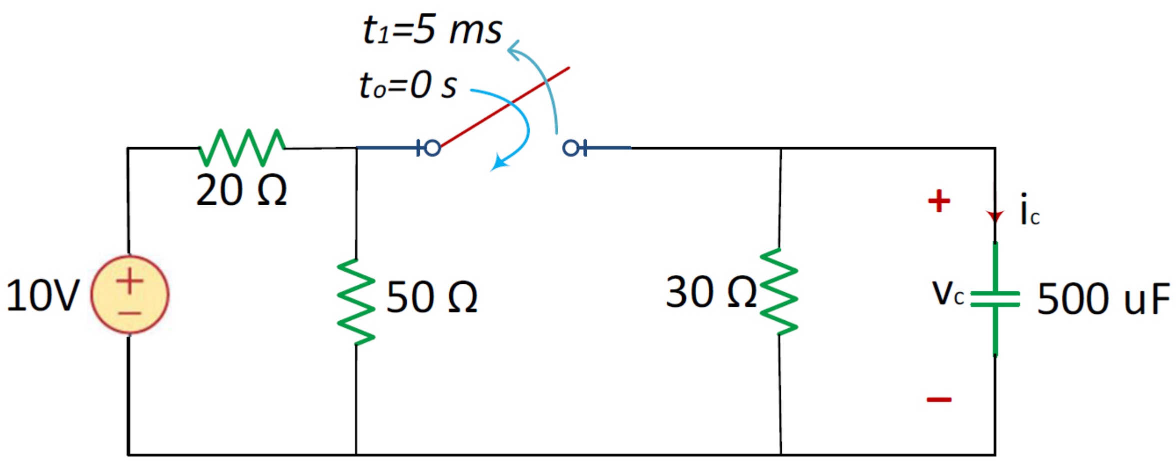


Figure 2

Answers:

i) $v_c(t)$ = 4.839 (1–$e^{-206.65t}$) V,

ii) $v_c(t)$ = 3.117$e^{-66.667(t-0.005)}$ V,

iii) i_c (t=10ms) = – 0.075 A,

iv) i_{20} = 0.5 – 0.242(1–$e^{-206.65t}$) A

Solution to the modified question:

The current flowing through the 20 Ω resistor for the time interval 0 ≤ t < 5 ms:

$$i_{20} = \frac{10 - V_c}{20k} = \frac{10 - 4.839(1 - e^{-206.65t})}{20} \text{ A}$$

$$= 0.5 - 0.242(1 - e^{-206.65t}) \text{ A}$$

Q3 (<u>Sample exam question</u>)

A phonograph pickup, stereo amplifier, and speaker are shown in Figure 3a and redrawn as a circuit model as shown in Figure 3b.

i) Calculate voltage v_x.

ii) Determine the resistance R so that the voltage v_o across the speaker is 16 V.

iii) Calculate the power in the dependent voltage source.

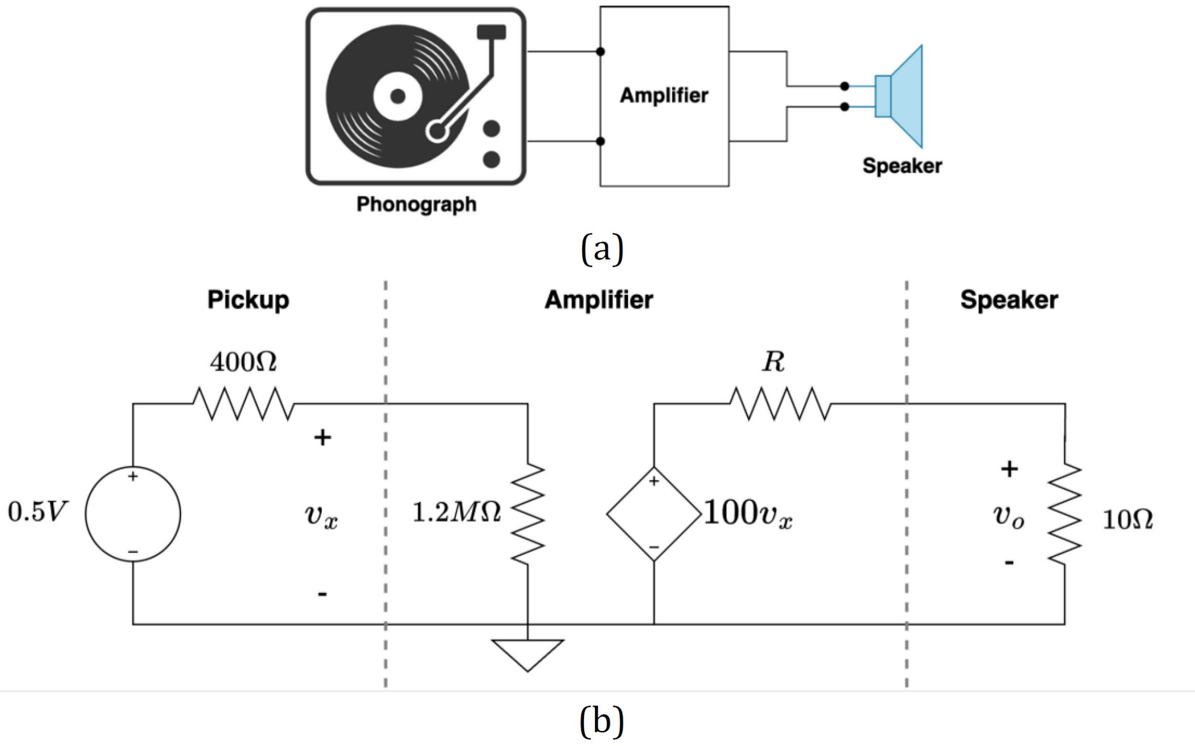
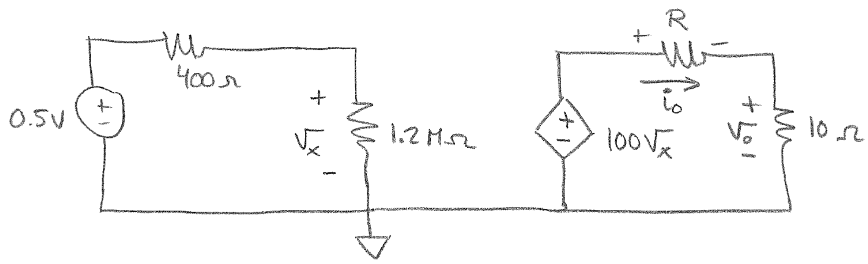


Figure 3

Answers: i) v_x = 0.5 V, ii) R = 21.25 Ω, iii) P = -80 W

Solutions:



i)

$V_o = 16V \Rightarrow i_o = \frac{V_o}{10} = \frac{16}{10} = 1.6\,A$

KVL@ right mesh: $-100V_x + Ri_o + 10i_o = 0$

where $V_x = 0.5 \frac{1.2M}{1.2M + 400\Omega} = 0.49983\,V \simeq 0.5V$

voltage divider

$-100 \times 0.5 + (10+R) \times 1.6 = 0$

$(10+R)\,1.6 = 50 \Rightarrow 10 + R = \frac{50}{1.6} = 31.25$

$\boxed{R = 31.25 - 10 = 21.25\,\Omega}$

ii)

$\boxed{P_{ds}} = -V_{ds} \times i_{ds} = -100V_x \cdot i_o =$

$= -100 \times 0.5 \times 1.6 = \boxed{-80\,W}$

Q4 (Sample exam question)

Determine V_0 and the power in the voltage source P_s for the circuit illustrated in Figure 4 under the following conditions:

i) When element X is a resistor with R=3kΩ.

ii) When element X is replaced by a short circuit.

iii) When element X is an open circuit.

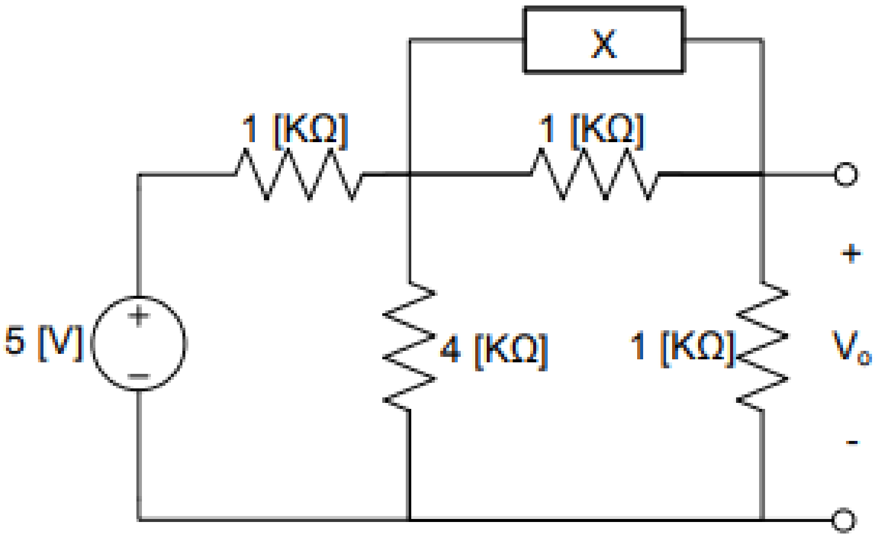
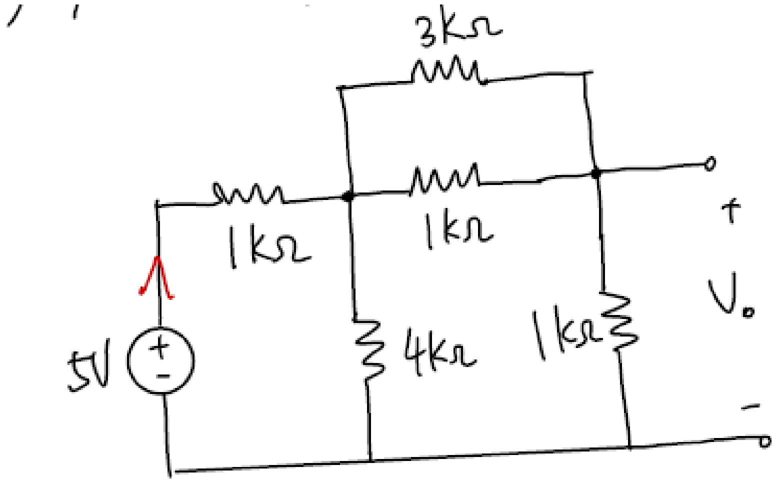


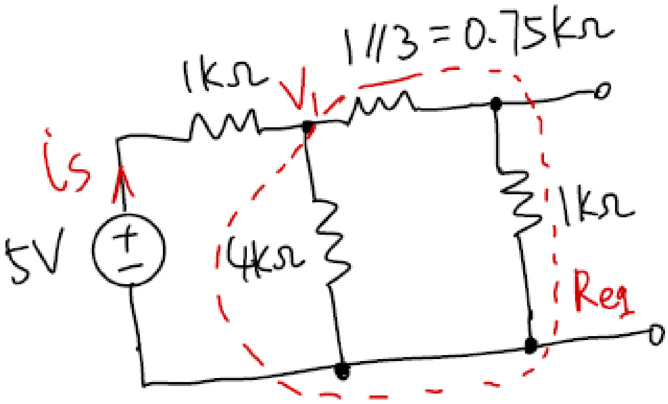
Figure 4

Answers: i) V_0 = 1.57 V, P_s = -11.26 mW, ii) V_0 = 2.22 V, P_s = -13.9 mW, iii) V_0 = 1.43 V, P_s = -10.72 mW

Solutions:

i) When element X is a resistor with R=3kΩ.





$$V_1 = 5 \times \frac{\overbrace{4k // (0.75k + 1k)}^{R_{eq}}}{1k + 4k // (0.75k + 1k)}$$

$$R_{eq} = \left(\frac{4 \times 1.75}{4 + 1.75}\right) k\Omega = 1.22 k\Omega$$

$$V_1 = 5 \times \frac{1.22 k\Omega}{2.22 k\Omega} = 2.748 \text{ V}$$

↑ voltage divider

$$V_o = 2.748 \times \frac{1k}{0.75k + 1k} = 2.748 \times \frac{1}{1.75} = 1.57 \text{ V}$$

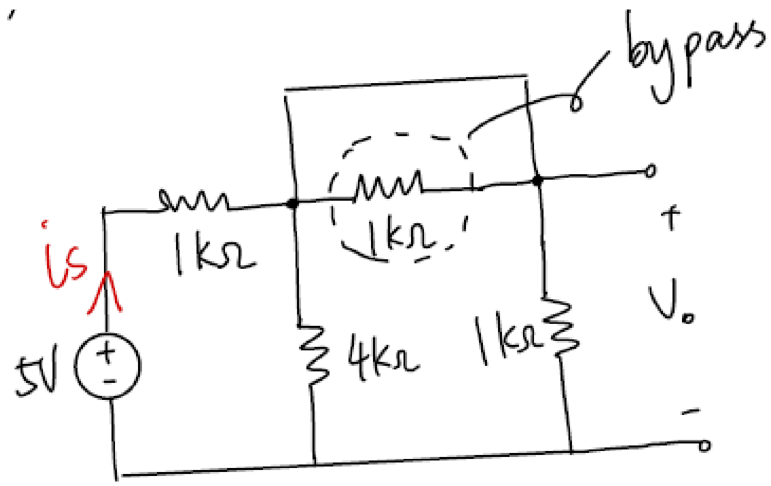
↑ voltage divider

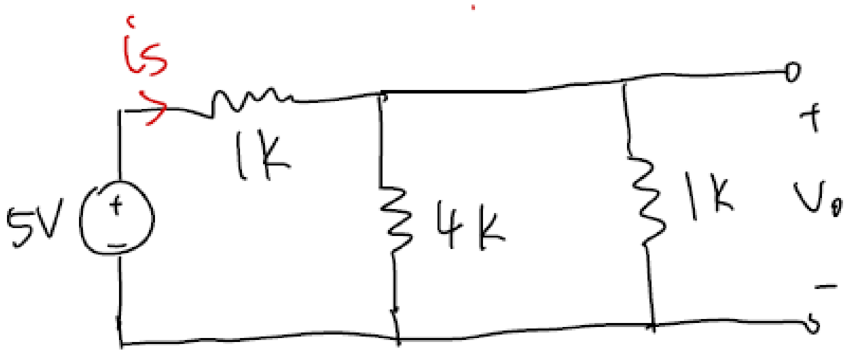
$$i_s = \frac{V_s - V_1}{1k} = \frac{5 - 2.748}{1000} = 2.252 \text{ mA}$$

$$P_s = -V_s i_s = -5 \times 2.252 = -11.26 \text{ mW}$$

↑ negative sign indicates "supplied"

ii) When element X is replaced by a short circuit.





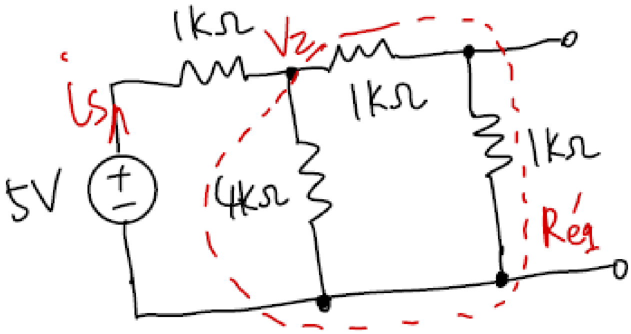
$$V_0 = 5 \times \frac{1K // 4K}{1K + 1K // 4K} = 5 \times \frac{0.8K}{1.8K} = 2.22\,V$$

voltage divider

$$i_s = \frac{V_s - V_o}{1K} = \frac{5 - 2.22}{1K} = 2.78\,mA.$$

$$P_s = -V_s i_s = -5 \times 2.78 = -13.9\,mW$$

iii) When element X is an open circuit.



$$V_1 = 5 \times \frac{\overbrace{4k /\!/ (1k+1k)}^{R'_{eq}}}{1k + 4k /\!/ (1k+1k)}$$

$$R'_{eq} = \left(\frac{4 \times 2}{4+2}\right) k\Omega = 1.333\Omega$$

$$V_2 = 5 \times \frac{1.333}{2.333} = 2.86V$$

voltage divider

$$V_0 = 2.86 \times \frac{1k}{1k + 1k} = 2.86 \times \frac{1}{2} = 1.43 \text{ V}$$

voltage divider

$$i_s = \frac{V_s}{R'_{eq} + 1k} = \frac{5}{2.333k} = 2.143 mA$$

$$P_s = -V_s i_s = -5 \times 2.143 = -10.72 \text{ mW}$$